CHITKARA UNIVERSITY

Ayush Bhrouta <ayush1421.be22@chitkara.edu.in>

3rd International Conference on Machine Learning Algorithms : Submission (9) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org> Sat, May 2, 2026 at 5:56 PM
To: ayush1421.be22@chitkara.edu.in

Hello,

The following submission has been edited.

Track Name: Track 2: Machine Learning for Healthcare and Computer Vision

Paper ID: 9

Paper Title: Diabetes Prediction Using the PIMA Indians Dataset

Abstract:
Abstract—
Early identification of diabetes is essential for effective disease management and reduction of long-term health complications. Machine learning techniques have increasingly been applied to support predictive analysis in healthcare; however, selecting an appropriate model requires balancing predictive performance and practical applicability.

This study presents a comparative evaluation of four supervised machine learning models—Logistic Regression, Decision Tree, Support Vector Machine (SVM), and Random Forest—for diabetes prediction using the PIMA Indians Diabetes Dataset. A consistent experimental framework is adopted, including preprocessing steps such as handling invalid values, feature normalization, and stratified train-test splitting to ensure reliable comparison across models.

The performance of each model is assessed using standard classification metrics, including accuracy, precision, recall, and F1-score, along with visual analysis through confusion matrices and ROC curves. Experimental results demonstrate that the Random Forest model achieves the highest accuracy of 91%, outperforming the other models, followed by SVM (86%), Decision Tree (83%), and Logistic Regression (79%).

The findings suggest that ensemble-based methods are more effective in capturing complex relationships in clinical data and improving prediction reliability. This work provides useful insights for selecting suitable machine learning models in healthcare applications and supports the development of decision-support systems for early diabetes detection.

Created on: Sun, 26 Apr 2026 13:19:15 GMT

Last Modified: Sat, 02 May 2026 12:26:49 GMT

Authors:
- ayush1421.be22@chitkara.edu.in (Primary)
- saksham768.be22@chitkara.edu.in
- rupindeep750.be22@chitkara.edu.in

Primary Subject Area: ML for Healthcare

Secondary Subject Areas: Not Entered

Submission Files:
Diabetes Prediction Using the PIMA Indians Dataset.pdf (646 Kb, Sat, 02 May 2026 12:26:07 GMT)

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